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1- SUMMARY OF PROCESS

1. Both computers must have a valid IP address.
2. Check the contents of the "hosts" file. Each computer should have its own IP address on one line and at least the IP address of the destination computer on another line.
3. WS-FTP Tool must be configured.

2- SETING UP THE PC TO HOST (SGI) NETWORK FOR WS_FTP

2-1 Creation of a Network

The 3 basic items necessary to establish a network are:

1. A unique IP address for the host (Octane) computer
2. A unique IP address for the PC.
3. A valid Netmask.

Usually, IP addresses are assigned to the Signa System during the system install process. To determine what addresses have already been assigned to the MR System open up the Install GUI and select the Network tab. It will tell you what addresses have already been assigned to the Host (Octane) the PC and what netmask is currently being used. Write these important numbers down for future reference. You will need them later in this procedure.

2-2 Identifying the Host (SGI) network configuration

1. From the Tools Menu, open the Guided Install/Reconfiguration GUI.



Select Guided Install/Reconfiguration
Password: operator

SERVICE DESKTOP TOOLS MENU

Illustration 2-1

1. Select the Network tab.

2. Host IP address to: 192.1.1.1

3. The Netmask .

4. IWS PC IP Address to: 192.1.1.2

5. IP Address for the Laptop port:

Note
Netmask is in Hexidecimal notation

0xfffffc00				
Place holder	0x	ff	ff	fc 00
		255	255	252 0

NETWORK INFO TAB
ILLUSTRATION 2-2

6. Write the numbers down.
7. Exit the Install/Reconfiguration GUI
8. Exit the Tools Menu

2-3 Setting up a Temporary Network

If you cannot determine what the legal IP addresses are, you may find it necessary to "make up" an IP address for both computers perform the following steps:

1. Make sure to isolate the MR system from any hospital/building network or the internet first. The *Network Hub* found in the Operators Workspace, is the safest place to DISCONNECT a hospital/building network or internet network connection from the MR system.
2. Open up the Guided Install/Reconfiguration GUI found under the Tools Menu.



Select Guided Install/Reconfiguration
Password: operator

SERVICE DESKTOP
Illustration 2-3

3. Select the Network tab.
4. Enter Host IP address to: 192.1.1.1
5. Enter IWS PC IP Address: 192.1.1.2

Note

The Netmask is already assigned and Cannot be changed.

6. Enter IP Address for the Laptop port:

Note

Netmask is in Hexidecimal notation

0x	ff	ff	fc	00
Place holder	255	255	252	0

NETWORK INFO TAB
ILLUSTRATION 2-4

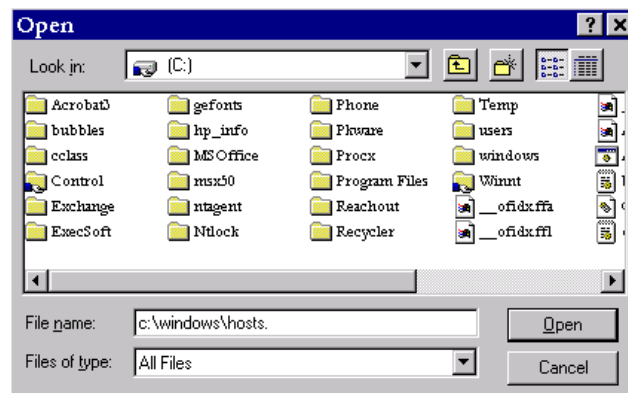
7. Exit the Install/Reconfiguration GUI
8. Exit the Tools Menu
9. Select Shutdown, Reboot the computer.

Note

Any changes to the "hosts" file of either the Octane or the PC must be followed by a complete reboot of the Octane or the PC to establish communication between them.

2-3 PC hosts File Configuration

1. To configure the local network host file select the Task Bar **[Start]** button. Move up and select **"Programs"** then **"Accessories"** then over to **"NotePad"**. The NotePad text editor will start. In NotePad, select **[File]** then **[Open]**.



WINDOWS NOTEPAD
ILLUSTRATION 2-5

2. Change the "File of type" box to "All Files(*.*)".

3. In the "File Name" box type:

c:\windows\hosts.

Note: The period after c:\windows\hosts
MUST be included in the "File Name" box.

- If the hosts file opens **Proceed with step 4:**
- If the hosts file IS NOT present, you will get an error: **Proceed to Section 2-4.**



ERROR
ILLUSTRATION 2-5

4. The host file will now open. Add the following new entries at the bottom of the hosts file.

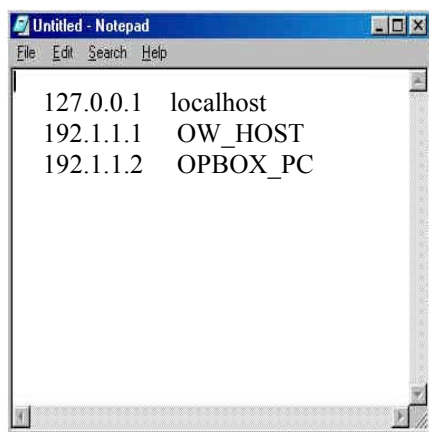
192.1.1.1 **OPBOX_PC** (OR should us the IP Address of this PC)
192.1.1.2 **OW_HOST** (OR should use the IP Address of the SGI Host)

In NotePad, select "File" then "Save" and "File" then "Exit" Note Pad.

5. Reboot the PC for the change to take affect.

2-4 Creating a PC 'hosts' file

1. To configure the local network host file select the Task Bar **[Start]** button. Move up and select "Programs" then "Accessories" then over to "NotePad". The NotePad text editor will start.

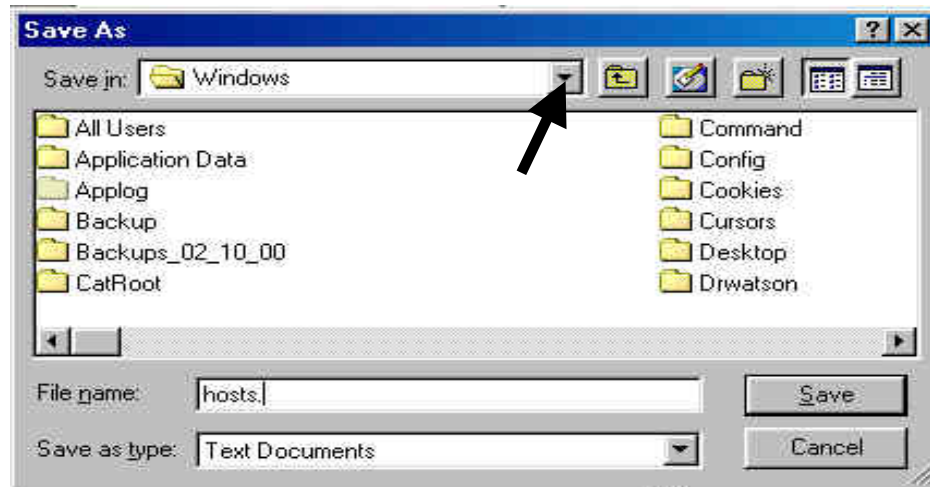


WINDOWS NOTEPAD
ILLUSTRATION 2-6

2. Create the hosts file by adding the following new entries as shown in Illustration 2-6.

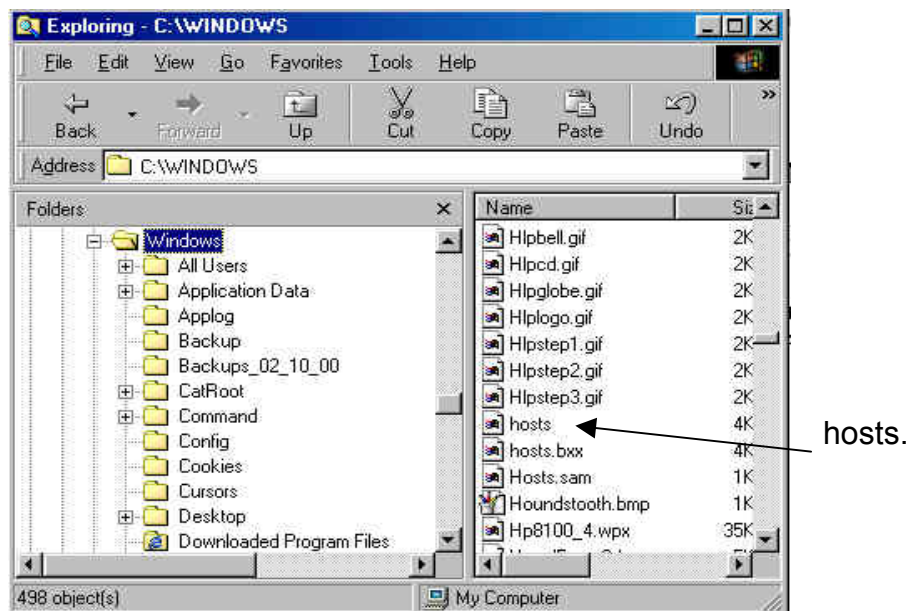
192.1.1.1 OW_HOST (Or use the IP Address of the
SGI Host)
192.1.1.2 OPBOX_PC (Or use the IP Address of this
PC)

3. IN NOTEPAD, SELECT **"FILE"** THEN **"SAVE AS"** .YOU WILL NEED TO CLICK ON THE FOLDERS UNTIL YOU GET TO THE WINDOWS DIRECTORY/FOLDER.



SAVE AS SCREEN
ILLUSTRATION 2-7

4. Once you are at the Windows directory/folder, type **"hosts."** As the filename. Ensure you don't use the quotes, however you **MUST** add the dot (".") after the name (hosts.)
5. **"Exit"** Note Pad.
6. Check to see if the new file is now in the Windows directory/folder, using Windows Explorer:



WINDOWS EXPLORER
ILLUSTRATION 2-8

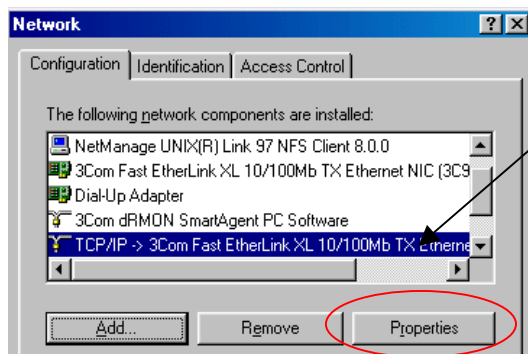
7. Reboot the PC for the change to take affect.

3- LOCATING/CHANGING THE PC NETWORK IP ADDRESS

This section describes the "typical" process to acquire network addresses assigned to the Signa PC and Field Laptops. In general, it should help you configure your system. Depending on age, the PC in the Signa system may have different (NIC) Network cards.

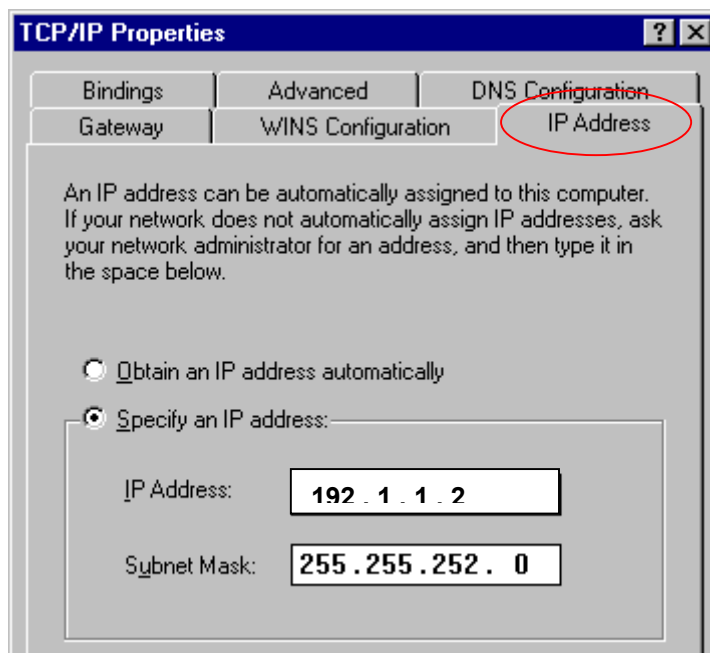


1. From the Windows Start Menu, open **[Settings]** and then **[Control Panel]**.
2. Locate and open the **[Network]** icon by double clicking with the left mouse button.



CONFIGURATION TAB
ILLUSTRATION 3-1

2. At the main Network box, select the **"TCP/IP -> 3Com....."** (Select your NIC Card) entry and select the **[Properties]** button.
3. Select the **[IP Address]** tab. Make sure the "Specify an IP Address" is selected. The PC IP address and subnet mask should be listed. Write them down.



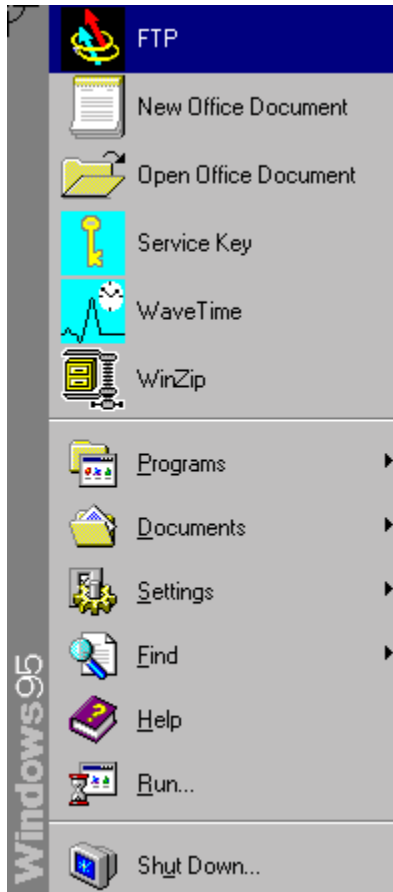
IP ADDRESS TAB
ILLUSTRATION 3-2

4. To Exit Select the **[OK]** button at the bottom of the "Properties" page, then the **[OK]** button at the bottom of the "Network" page.
5. Close Control Panel by clicking on the **X** in the upper right corner of the window.
6. If changes were made a pop-up window appears: select **[YES]** to reboot the PC for changes to take effect.

4- WS-FTP95 PRO (FTP)TOOL FOR WINDOWS 95

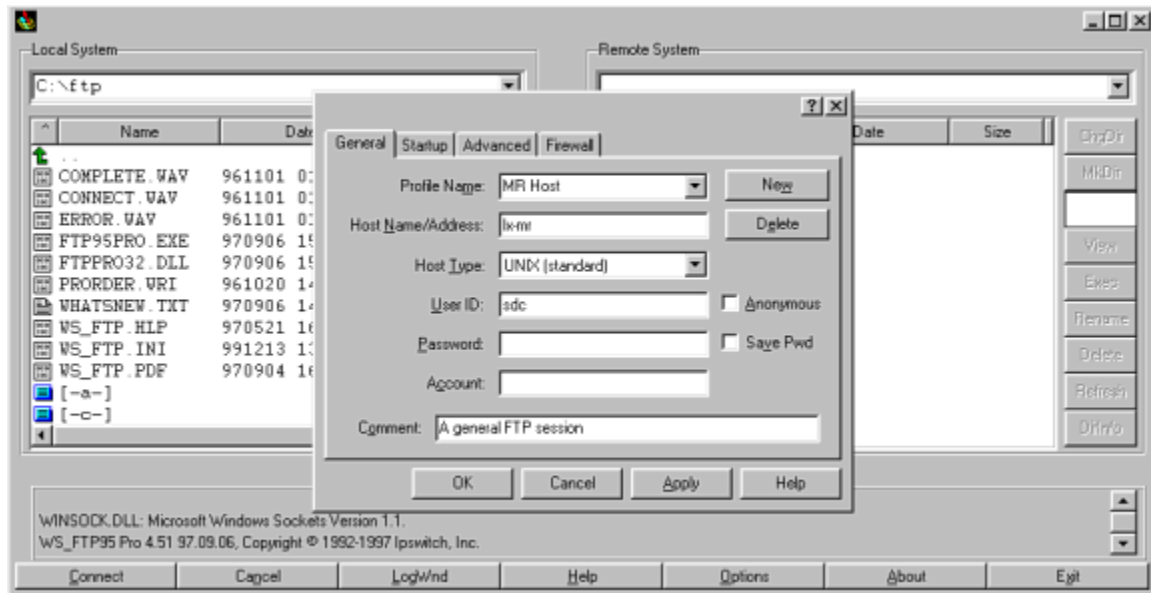
The standard PC load comes configured with a FTP tool to make file transfers between the system PC and the host computer easier to do. The tool can be used to transfer images, raw data, test files or any other text or binary files to or from either computer.

To start the tool, select on the task bar, the [Start] button then scroll to ftp. See Illustration 5-1.



STARTING THE FTP TOOL
ILLUSTRATION 4-1

The tool comes pre-configured for usual transfers made between the computers. See Illustration 4-2.

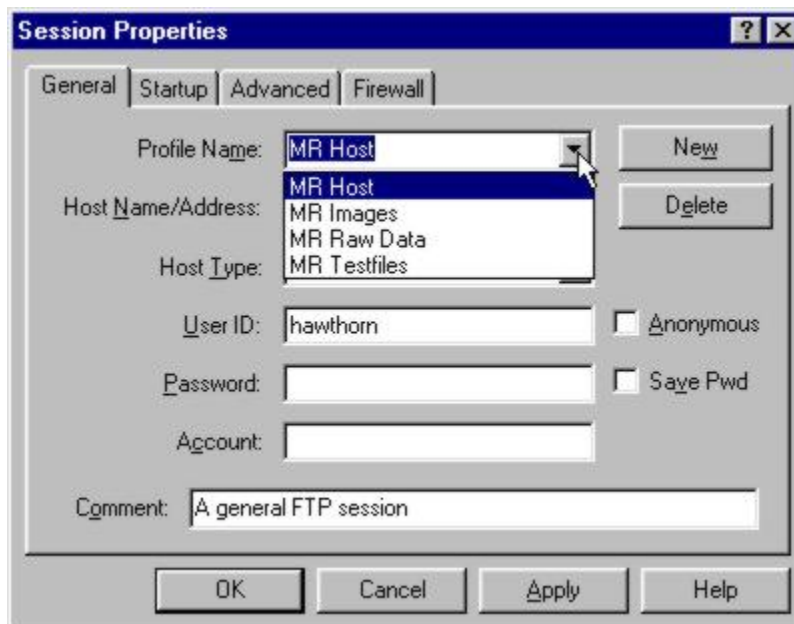


INITIAL STARTUP OF THE FTP TOOL
ILLUSTRATION 4-2

Note

Use of this tool as configured assumes the PC's host file has been updated with the host IP address. See Section 2 and 3, to update the host file if needed.

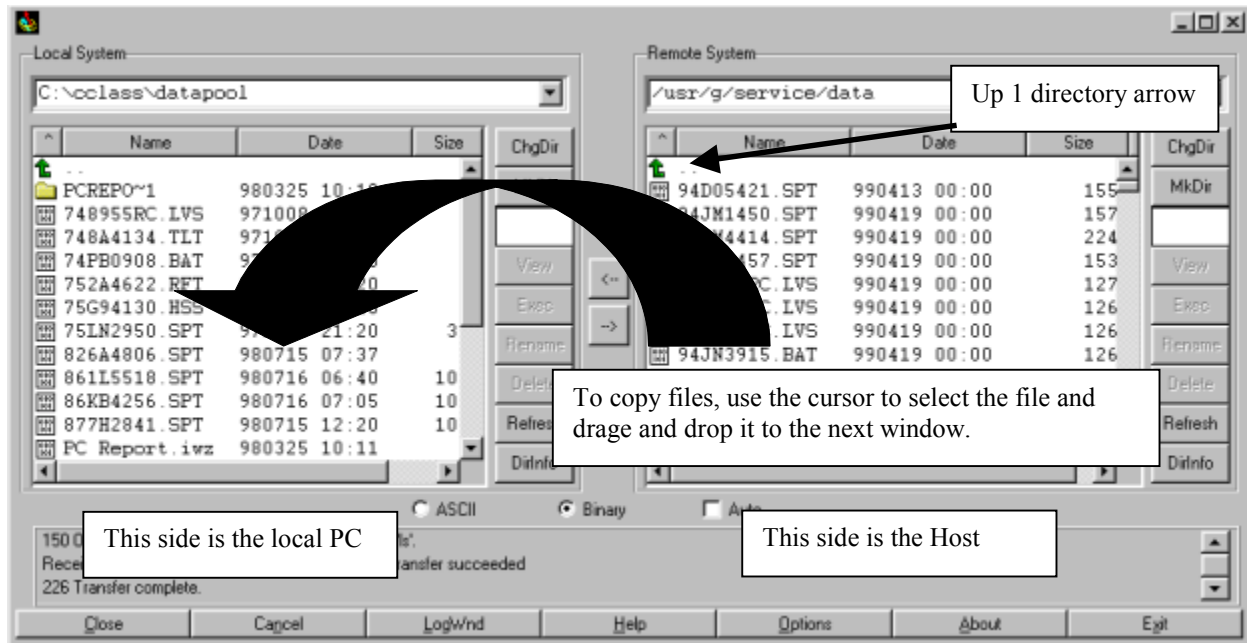
Several locations have been pre-loaded into the FTP tool. See illustration 4-3.



SESSION SELECTION
ILLUSTRATION 4-3

To transfer files from the host to the PC;

1. Select a session that would put you closest to the desired data.
2. Enter a User Name. The Default would be [sdc]. This is case sensitive, user lower case.
3. Enter the systems password. Signa default for sdc would be **adw2.0 <Enter>**.



THE FTP USER INTERFACE
ILLUSTRATION 4-4

For further discussion on the FTP tool itself and the "how to's" of using the tool, press the **[Help]** button at the bottom of the interface.

5- WS-FTP98 PRO (FTP)TOOL FOR WINDOWS 98

5-1 Overview

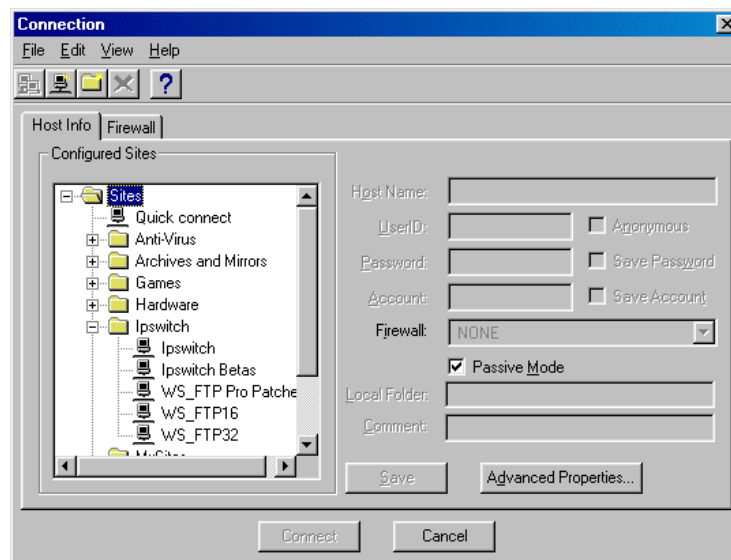
All LX Signa Systems have this FTP software on the PC. This FTP Tool was selected because of its simple setup and usage.

5-2 WS_FTP Pro Configuration

1. Double click WS_FTP Pro icon or select WS_FTP Pro from startup menu.

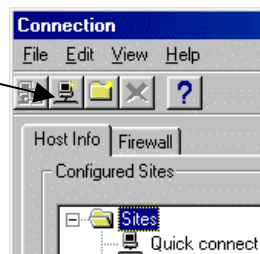


WS_FTP PRO ICON
ILLUSTRATION 5-1



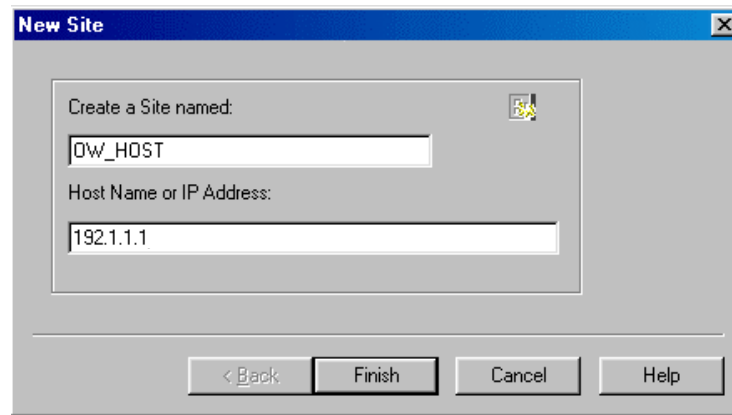
CONNECTION WINDOW
ILLUSTRATION 5-2

2. Create New Site:



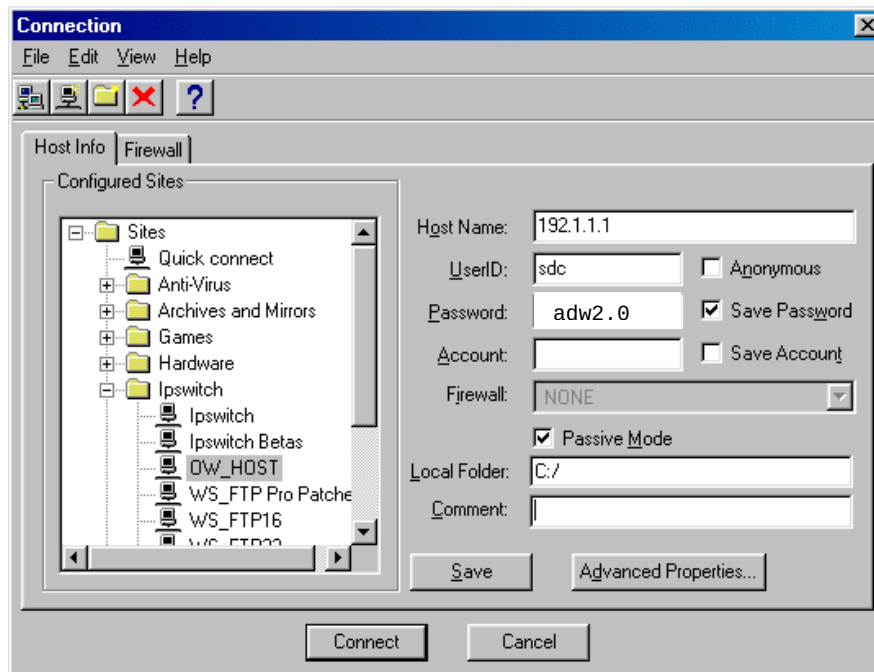
NEW SITE
ILLUSTRATION 5-2

3. Enter Site Name. (e.g. signa)
4. Enter IP Address assigned to the PC.



NEW SITE
ILLUSTRATION 5-3

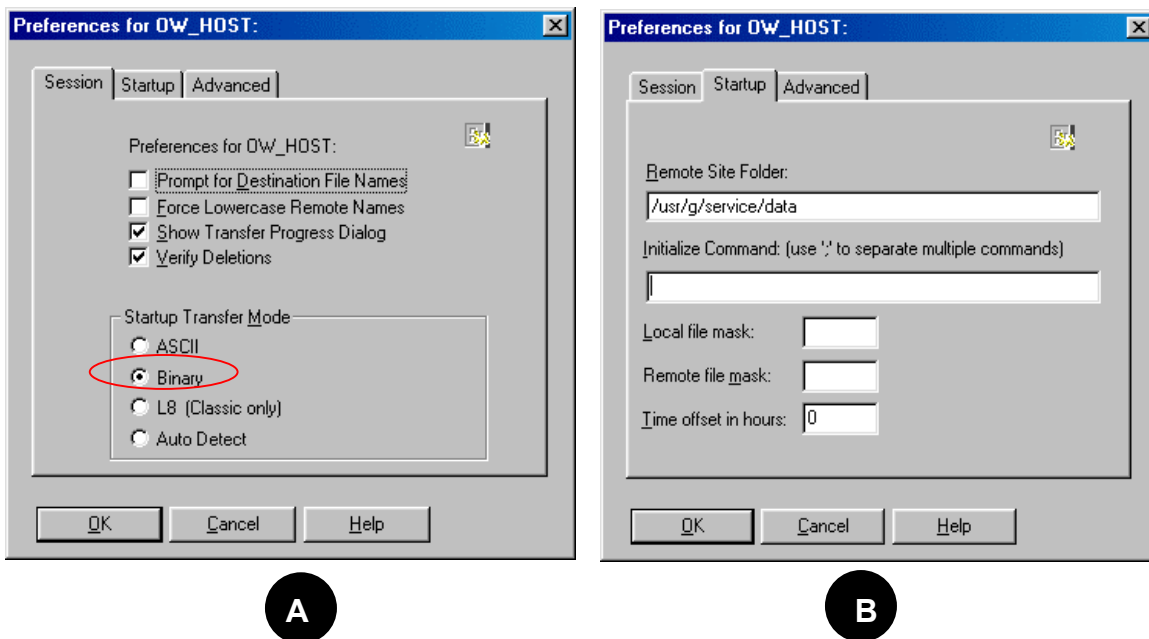
5. Check that the new icon is created in configure site window.



CONNECT WINDOW WITH NEW SITE
ILLUSTRATION 5-4

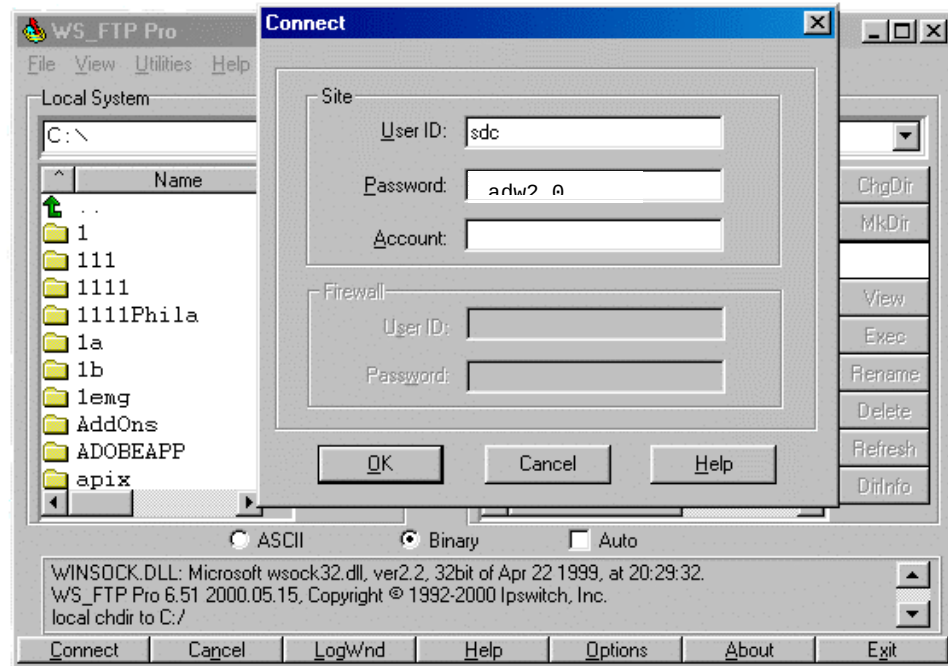
6. Refer to Illustration 9-4. Enter the Host Name as shown : **sdc**
7. Refer to Illustration 9-4. Enter the password: **adw2.0** (it will show up as *****).
8. In the Local Folder boxType: **c:/** (Any data to the PC will show up in the C:/ directory of the PC)

9. Click [Advanced Properties] to setup Preferences.



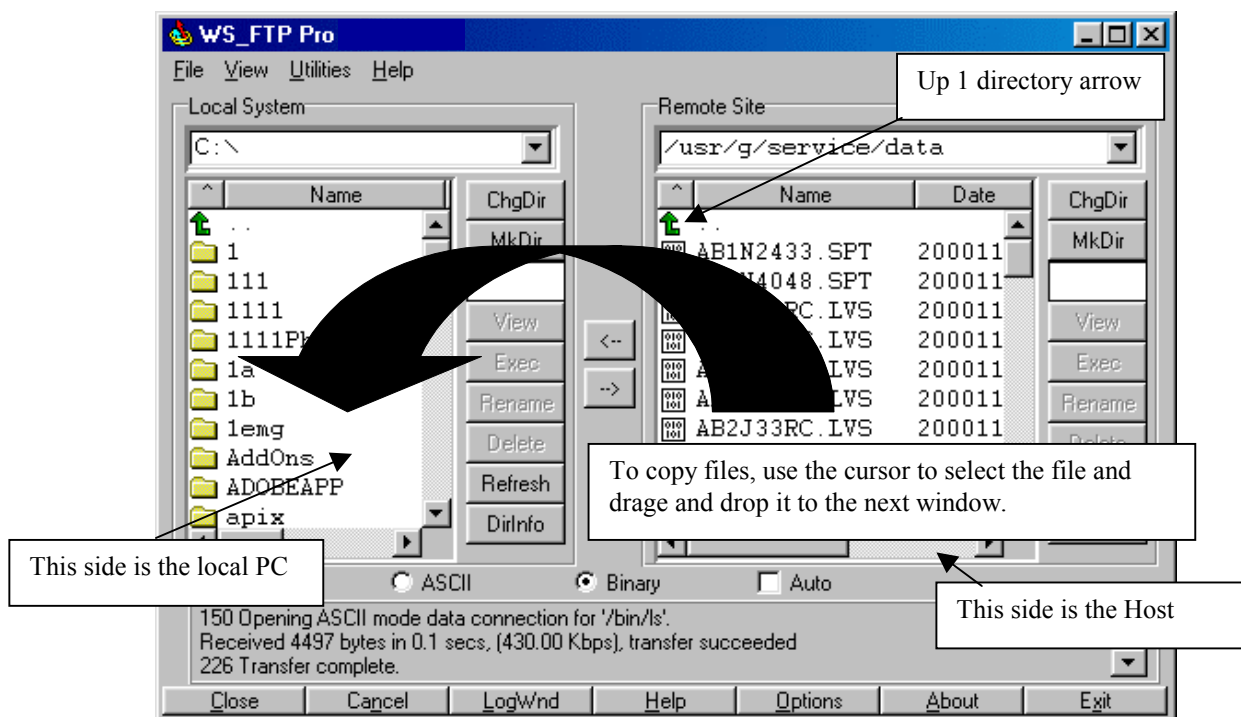
PREFERENCES
ILLUSTRATION 5-5

10. Setup the session Tab like Illustration 5-5 A and the Startup Tab like illustration 5-5 B.
- Select StartupTransfer mode - **Binary**
 - Preferences for OW_HOST: ✓Check Show Transfer Progress dialog and ✓Verify deletions.
 - Change to Startup Tab. Type: **/usr/g/service/data** as the Remote Site Folder the tool will startup in.
11. Click [OK] The Preferences window will dissappear.
12. Click [Save].



CONNECT GUI
ILLUSTRATION 5-6

12. Enter password: **adw2.0** (Password will appear as *****)
13. Click [**Connect**].
14. Verify Host computer is connected as following illustration.



WS_FTP PRO 98 GUI
ILLUSTRATION 5-7

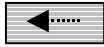
6- TRANSFERRING FILES WITH WS_FTP PRO 98

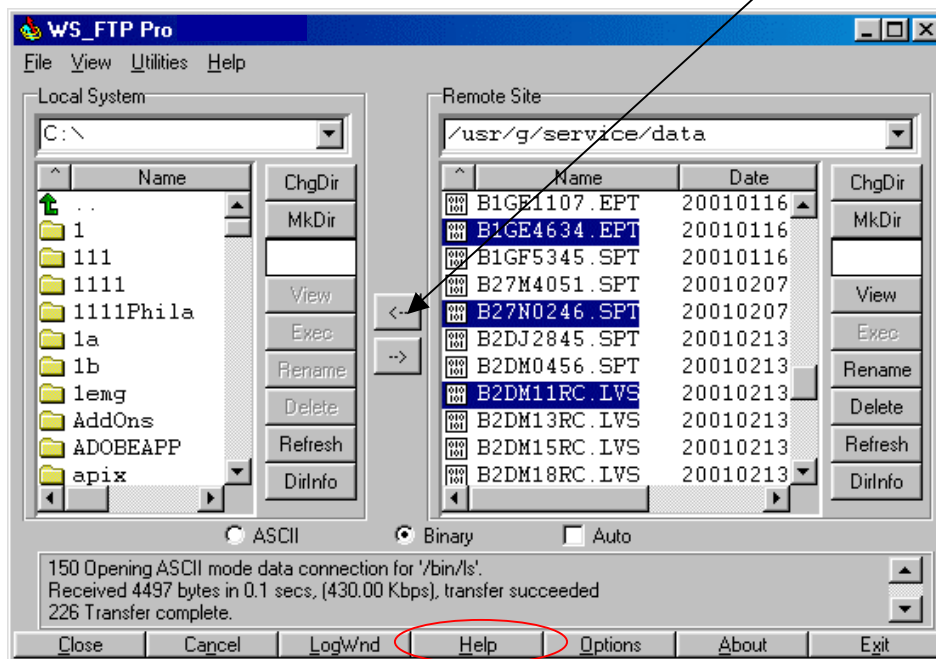
6-1 Transferring Files

There are several ways to transfer files. The simplest way to transfer a file is to select, "Drag & Drop" from one FTP window to the next.

1. To transfer a file **from OW_HOST to the OPBOX_PC** select the file and "Drag & Drop" it to the Local System Window.
2. To transfer a file **from OPBOX_PC to OW_HOST** select the file and "Drag & Drop" it to the Remote Site Window.

Other Methods:

- a. Select the file in the Remote Site window with one click of the mouse.
- b. To select a RANGE of files select the first file you want, then hold down the [ALT] + click the last file. the entire range of files will be selected.
- c. To select individual files, hold down the [CTRL] + filename for each file selected.
- d. Once the files needed are selected use the transfer button  Between the Local System Window and the Remote Site window.



Selected Files

Movable Window
Resize bar

File transfer type:
All executables
should be
transferred
"Binary".
Text Files can be
transferred ASCII.
When in doubt,
use BINARY.

TRANSFERRING FILES
ILLUSTRATION 6-1

3. To exit, Select File and from the Pull-down box select, **Exit**.

For further information on how to use the WS_FTP PRO 98 Interface use the Help button.

A- APPENDIX A

NOTE

IRIX is a Proprietary version of UNIX written by Silicon Graphics, Inc. For the purpose of discussion in this document, the difference between the two is not important.

To communicate between the any two computers, each computer must be aware of the other's address. This can be done by using a file called "hosts". Because communication between computers may cross Operating Systems such as UNIX,(IRIX) and DOS, it is extremely important to use *lower case* in naming files that would typically be moved between these two types of Operating Systems. The "hosts" filename itself should always be lower case on a UNIX based Operating System. There is no exception to this rule. The content of the "hosts" file is case sensitive as well.

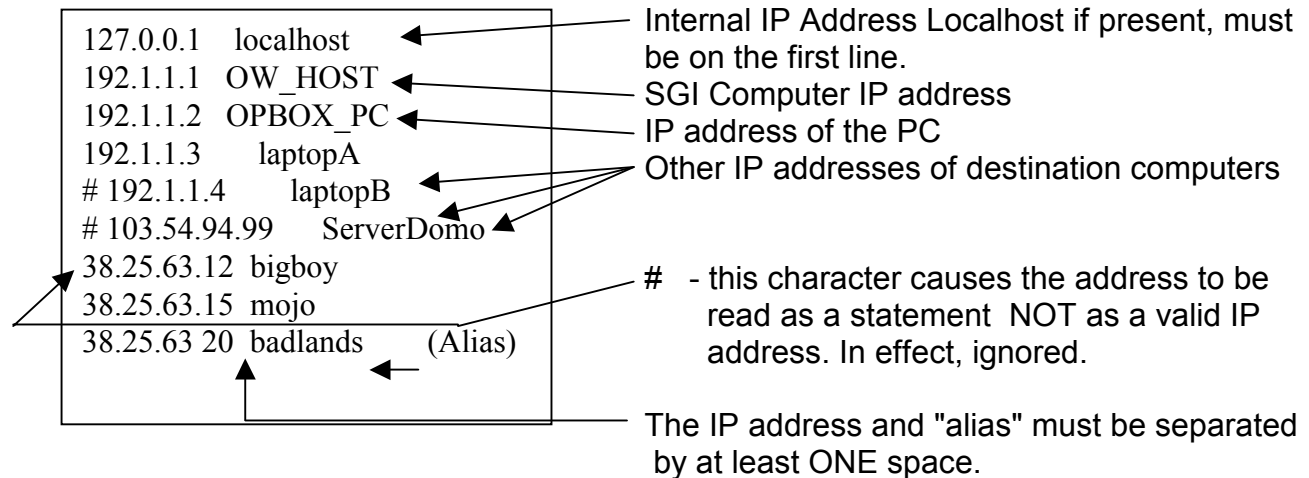
NOTE

Windows 95/98/ME all have a tendency to force a capital letter in Windows Explorer. Since the Windows Operating System is not case sensitive, the "hosts" file starts with an upper case "H", and will appear in the Explorer window as "Hosts". This is a known Windows issue, however, the rest of the industry has designed their software tools to ignore the upper case H. Also, and all contents of this special file will be case sensitive across platforms and Operating Systems.

The UNIX Operating System requires the "hosts" file to be in a special directory (/etc). It must be in this directory off of root (/) . All network communications depend on this file being in this location, /etc.

The "hosts" file in the Windows 95/98/ME Operating System is always in the /windows directory. (/Windows). Windows 95/98/ME all contain a sample hosts file as well. It is always in the /Windows directory as well. It is called Hosts.sam.

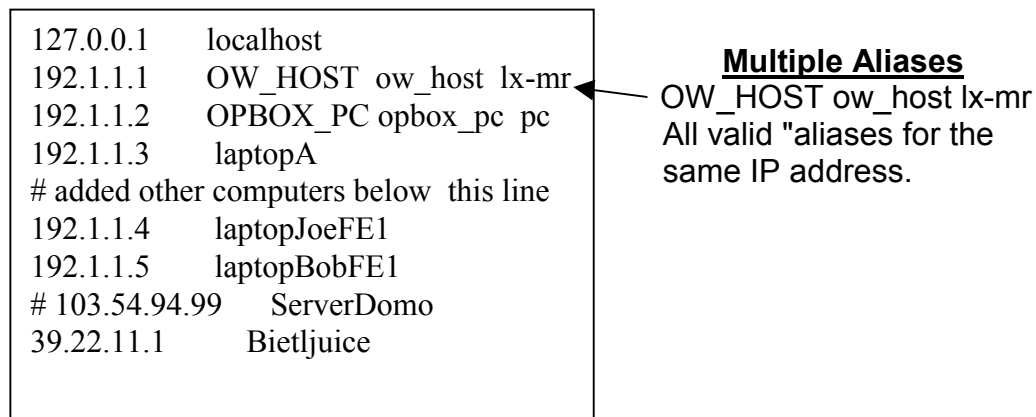
The last important fact about the "hosts" file is the importance of its filename. The UNIX operating System requires only the five characters "hosts" (Without quotes). The Windows Operating system typically follows an "8.3" standard for filenames. (Although Long File Name is supported as well). "8.3" means at least 8 characters followed by 3 characters, separated by a dot. or period ("."). When you create a file in Windows it requires you to at least include a dot or period at the end of the filename (".") Therefore when using Notepad, when you create a "hosts" file it requires you to save it as "hosts." The dot or period (".") is ignored by other Operating Systems.



TYPICAL hosts FILE
ILLUSTRATION A-1

About the alias name:

Alias names are "short cuts" that allow us to use a name for a computer versus the IP address. Aliases are usually easier to remember. An Alias is NOT required in the hosts file, it is placed there as a convenience to humans. In the example 192.1.1.1 is OW_HOST either "name" or IP number can be used as the Host ID for the Octane. You may also have more than one "name" associated with an IP address. Illustration 3-2 shows an upper case and a lower case name, and alternate names on the *same* IP address separated by at least ONE space.. This is to make it convenient for multiple users.



VALID HOSTS FILE
ILLUSTRATION A-2

About IP address 192.1.1.1

(Refer to illustration A-1 or A-2) This default IP address has been assigned at software load to be a "placeholder" until the Hospital/Site Network Administrator assigns a valid IP address, legal for the site.

About IP Addressing

Internet Protocol (IP) Addressing is broke into three main classes Class A, B and Class C. IP addresses are broken down into 4 Octets: 3.234.5.6

- **Class A:** First Octet is between :**1-127** (inclusive)
- **Class B:** First Octet is between :**128-191** (inclusive)
- **Class C:** First Octet is between :**192-223** (inclusive)

About IP address 192.1.1.1

(Refer to illustration A-1) This is a Class C address, and is assigned during the Host (Octane) software load procedure. It should be changed to a valid IP address issued by the hospital, clinic, office Network System Administrator. A minimum of two unique IP addresses must be acquired.

before you can configure a network between the host (octane) and the PC. The Host (Octane) will be assigned a unique address and the PC will also need a unique address. If at all possible, a third IP address should be assigned to the 10BaseT (RG45) jack on the outside of the Operators Workspace. This would typically be used for FE troubleshooting via a laptop.

About Subnet- Mask or "Netmask"

A *Netmask* allows for the creation of isolated networks or a *Sub-net*. A netmask is a series of bits designed to 'mask' certain portions of an IP address. The standard netmask for a class C network is 255.255.255.0 - basically the last octet, .0, says 'look here for the machine number, the rest (255) is network number'. The primary use for netmasks is in subnetting.

Subnetting a Class C Address: 223.1.2.0

To understand how to subnet, you have to step back and think a little about how routers make routing decisions and how they relate to the subnet mask. Further, you have to understand what composes an IP address and how a subnet mask divides the address up.

Let's take a typical Class C address network like **223.1.2.0** and break it down a little. All IP addresses are composed of 4 byte values or *octets*. The binary form of this address would look like: **11011111 . 00000001 . 00000010 . 00000000**

By applying a mask of **255.255.255.0** or:

```

      ← Network Number → | ← node →
      =====
    11111111 . 11111111 . 11111111 . 00000000
      255      .   255      .   255      .   0
  
```

You define which part of the address is the *network* and which part is the *node*. A Node is any device used to communicate on the network, such as a computer (Host). The generic case is **255.255.255.0** where the *node* address breaks on the last byte boundary. Thus, this Class C address would contain nodes in the range from 1 to 254, utilizing the full space of the last

octet. Each network must have a Net number and broadcast address, that's why you exclude 0 and 255 as nodes.

Note:

If you do not know how to create a "host" file on the PC refer to APPENDIX - A

- If the MR system is to be connected to a hospital/building Network or connected to the internet, you must ask your customer or the Network Administrator for these unique IP addresses.
- If the MR system is not on a hospital/building Network or connected to the internet you can temporarily assign valid IP addresses to the host (Octane) computer and the PC. Ask your customer or the Network Administrator for two addresses and a netmask number. The IP addresses must be ADDED to both hosts files on both systems.

IMPORTANT:

If the system is attached to a hospital/building Network or connected to the internet with temporary IP addresses, it is violation of Federal, State and International laws. You MUST have valid IP addresses legally assigned by a Network System Administrator.

Question: What if you don't know what the IP numbers are, and know one else knows either:

If you find it necessary to "make up" an IP address for both computers, make sure to isolate the MR system from any hospital/building network or the internet first.

The *Network Hub* found in the Operators Workspace. This is the safest place to DISCONNECT a hospital/building network or internet network connection from the MR system. Identify which cable it is, and temporarily disconnect it.

You can then edit both "hosts" files accordingly, accomplish the networking task between the host Octane and the PC, and then RESTORE the hosts files to their original configuration before reconnecting the system to the hospital/building network or the internet.

Any changes to the "hosts" file must be followed by a complete reboot of the Octane AND the PC to take affect.

WARNING:

Failing to restore the hosts files to their original condition will cause severe network problems if the MR system is attached to the hospital/building network or internet. It is also illegal to use an unofficially assigned IP address. Even if done out of accident this is a crime and carries heavy penalties and fines.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Jun, 26, 2001	D. Hofstetter	Initial version (removed from Host software install procedure).